

CCP (Cirrus Cloud Platform)

Whitepaper

Building a Sovereign Cloud Platform with Cirrus Cloud Platform

A unified cloud management platform for government and enterprise cloud services delivery

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Executive Summary

Governments and enterprises across the globe are accelerating their transition to sovereign cloud infrastructure — cloud platforms that are locally controlled, independently operated, and built to meet the unique compliance, security, and governance requirements of national and regulated industries. Cirrus Cloud Platform (CCP), developed by CoreEdge, is a hyper-scaler grade Cloud Management Platform (CMP) purpose-built to address this need. In combination with Cirrus Cloud Platform (CCP), the IaaS Orchestrator, and Cloud Orbiter, the Kubernetes Orchestrator, CCP delivers a unified cloud services platform that enables service providers

to offer Infrastructure-as-a-Service (IaaS), Platform-as-a-Service (PaaS), and Software-as-a-Service (SaaS) to their customers through a single, self-service portal. This White Paper presents the capabilities, architecture, service portfolio, and delivery approach of Cirrus Cloud Platform as deployed for a sovereign cloud platform serving government and enterprise customers. It is intended for technology leaders, cloud architects, and business decision makers evaluating a Cloud Management Platform for sovereign or enterprise cloud deployments.



KEY INSIGHT

Cirrus Cloud Platform provides hyper-scaler grade self-service capabilities for Infrastructure, Platform, and Software services — designed for sovereign cloud environments.

Key Highlights



Unified Cloud Management

Platform for IaaS, PaaS, and SaaS delivery



Self-service portal with **automated provisioning** and policy-based governance



Multi-region, multi-AZ high availability architecture with active-passive failover



Comprehensive service portfolio delivered in three phased milestones (MVP1, MVP2, MVP3)



Built-in Identity and **Access Management** with **multi-tenant identity** federation



Scalable to 50,000 virtual machines and **200,000 pods** per deployment



Enterprise-grade security with mTLS, AES-256 encryption, and RBAC

The Business Challenge

As cloud adoption matures, governments and enterprises are increasingly recognizing that public hyperscale cloud alone does not meet all their needs. Sovereign cloud — cloud infrastructure that is locally deployed, nationally controlled, and operated independently — is emerging as a critical requirement for sectors handling sensitive data, regulated workloads, and strategic national infrastructure. Building and operating a sovereign cloud platform at scale presents significant challenges:



Self-Service at Scale

Government and enterprise customers expect on-demand provisioning of compute, storage, and network resources without manual intervention from the service provider.



Visibility & Governance

Operating a multi-tenant cloud requires centralized visibility across all environments, with policy-based governance and compliance enforcement.



Operational Complexity

Managing heterogeneous infrastructure — physical servers, virtual machines, Kubernetes clusters, and storage — through disparate tools creates operational overhead and increases risk.



Security & Compliance

Sovereign cloud workloads demand end-to-end encryption, identity federation, and role-based access control that meets national and industry compliance standards.



Cost Optimization

Without centralized metering, showback, and quota management, cloud spending becomes opaque and difficult to govern across multiple tenants and business units.

The Solution — Cirrus Cloud Platform

Cirrus Cloud Platform (CCP) is a Cloud Management Platform developed by CoreEdge that provides hyper-scaler grade self-service capabilities for Infrastructure, Platform, and Software services. CCP is designed to operate as the management and orchestration layer above existing

cloud infrastructure, enabling service providers to deliver cloud services to their customers through a unified, self-service experience. CCP operates in combination with two complementary CoreEdge products:

Component	Role
Cirrus Cloud Platform (CCP)	Cloud Management Platform — the top-layer self-service and governance platform that customers and operators interact with directly.
Cirrus Cloud Platform (CCP)	IaaS Orchestrator — manages the underlying OpenStack infrastructure, providing compute, storage, and networking resources.
Cloud Orbiter	Kubernetes Orchestrator — manages Kubernetes clusters across the platform, enabling container workload management and application deployment.

Cirrus Cloud Platform - Platform Stack Architecture

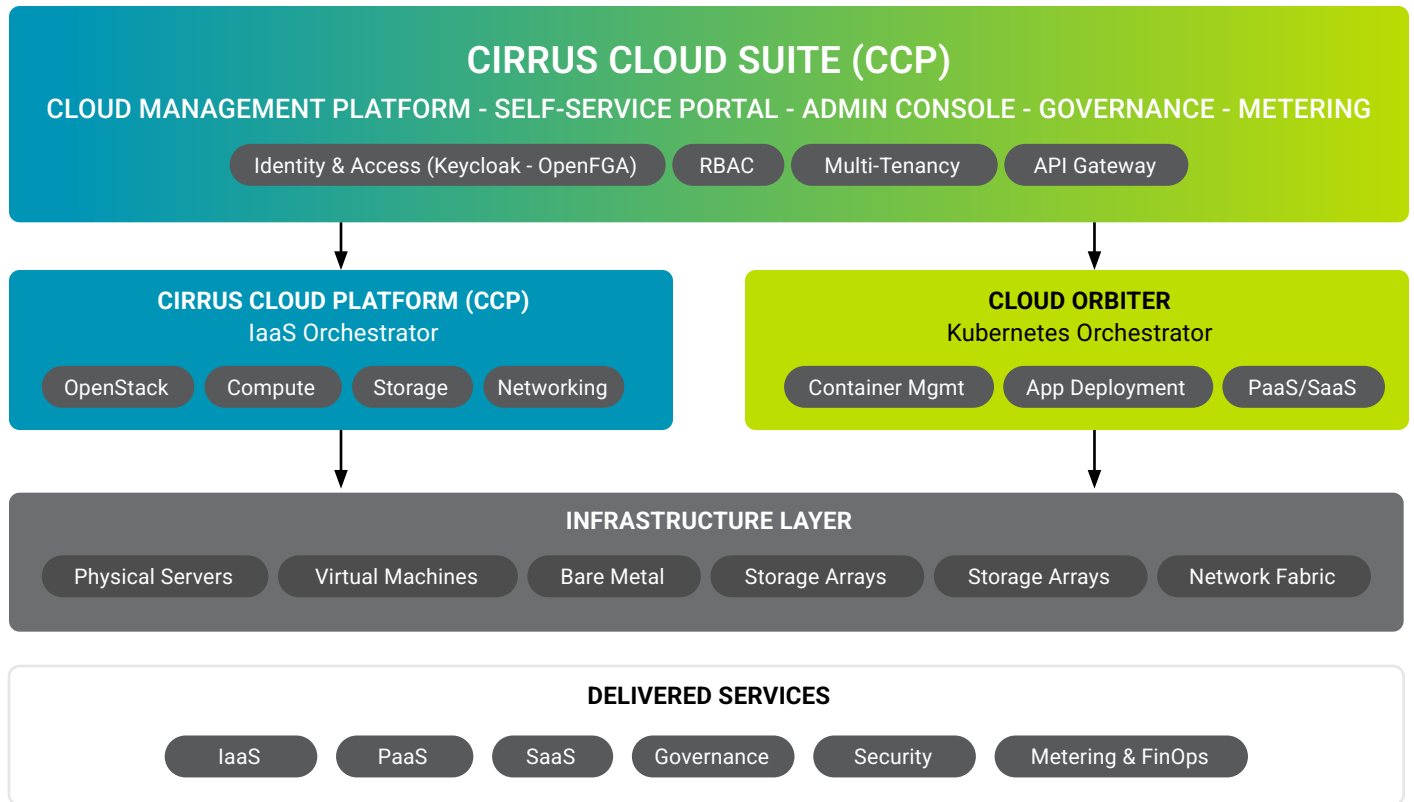


Figure 1: Cirrus Cloud Platform – Layered Platform Architecture

Together, these three components provide a complete, end-to-end sovereign cloud platform layer that serves both the internal teams of the cloud service provider – Day 2 operations, business units, security, FinOps, and cloud governance – and the end customers who consume cloud services.

Platform Capabilities

Cirrus Cloud Platform delivers five foundational capabilities that address the core challenges of sovereign and enterprise cloud management:

Five Core Platform Capabilities

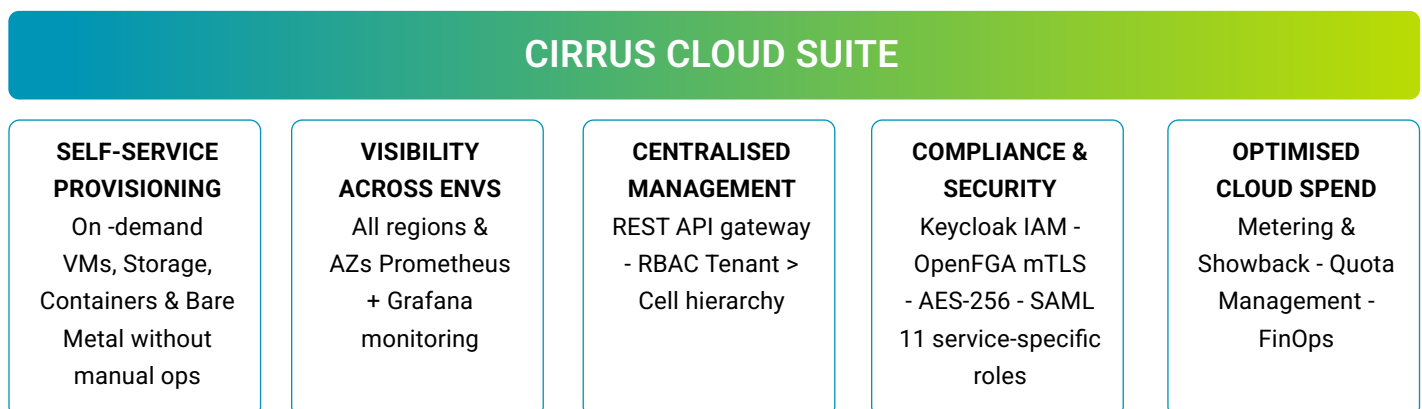


Figure 2: Five Core Platform Capabilities of Cirrus Cloud Platform



Self-Service Access for Automated Provisioning and Deployments

CCP provides a rich, user-friendly Self-Service Console as the primary interface for end users. Through intuitive interfaces, customers can provision and manage virtual machines, storage volumes, load balancers, container workloads, and bare metal servers without requiring intervention

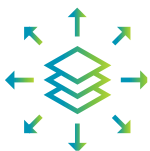
from the service provider's operations team. Organisation administrators can create and manage Projects and Cells, define access control policies, and ensure proper resource allocation and usage — all from a single portal.



Visibility Across Environments

The CCP Admin Console provides the service provider's operations team with a comprehensive view of the entire cloud environment — all virtual machines, volumes, load balancers, container namespaces, and infrastructure health indicators — across all regions and availability zones.

Integrated monitoring through Prometheus and Grafana enables proactive health tracking, alerting, and automated recovery workflows for cluster and database components.



Centralised Management

CCP Coredge Platform Services is composed of purpose-built microservices that communicate through well-defined REST APIs and internal routing mechanisms. The platform provides centralized access control and API logging through an in-built API gateway, ensuring secure and

authorized access to all platform resources from a single management plane. The platform's resource hierarchy — Tenant > Cell > Resources — provides a structured, governed model for multi-tenant resource management, with quota management enforced at both tenant and cell levels.



Improved Compliance and Security

CCP provides multi-layered security architecture. The built-in Identity and Access Management server (powered by Keycloak) is multi-tenant by design, with the capability to federate with external identity providers including BSS Portal and ADFS using SAML 2.0. For each customer organisation, CCP creates a unique identity account, ensuring complete isolation between tenants. Role-Based Access Control (RBAC) is enforced across all

platform layers through OpenFGA, Coredge's AuthZ engine. Eleven pre-defined service-specific roles and seven organisation-level roles provide granular access control aligned with the principle of least privilege. All data is protected with encryption in transit (mTLS) and encryption at rest (AES-256). Regular security assessments and compliance checks are built into the platform's operational model.



Optimised Cloud Spends

CCP includes built-in metering, showback, and quota management through the orbiter-metering module. This provides full visibility in resource consumption across all tenants and cells, enabling

the service provider to generate notional invoices, enforce spending limits, and give customers visibility into their own cloud usage.

Service Portfolio

CCP delivers cloud services to customers in a phased manner across three milestones. This approach ensures that foundational services are delivered and stabilised before expanding the portfolio, reducing deployment risk and enabling early value realisation.

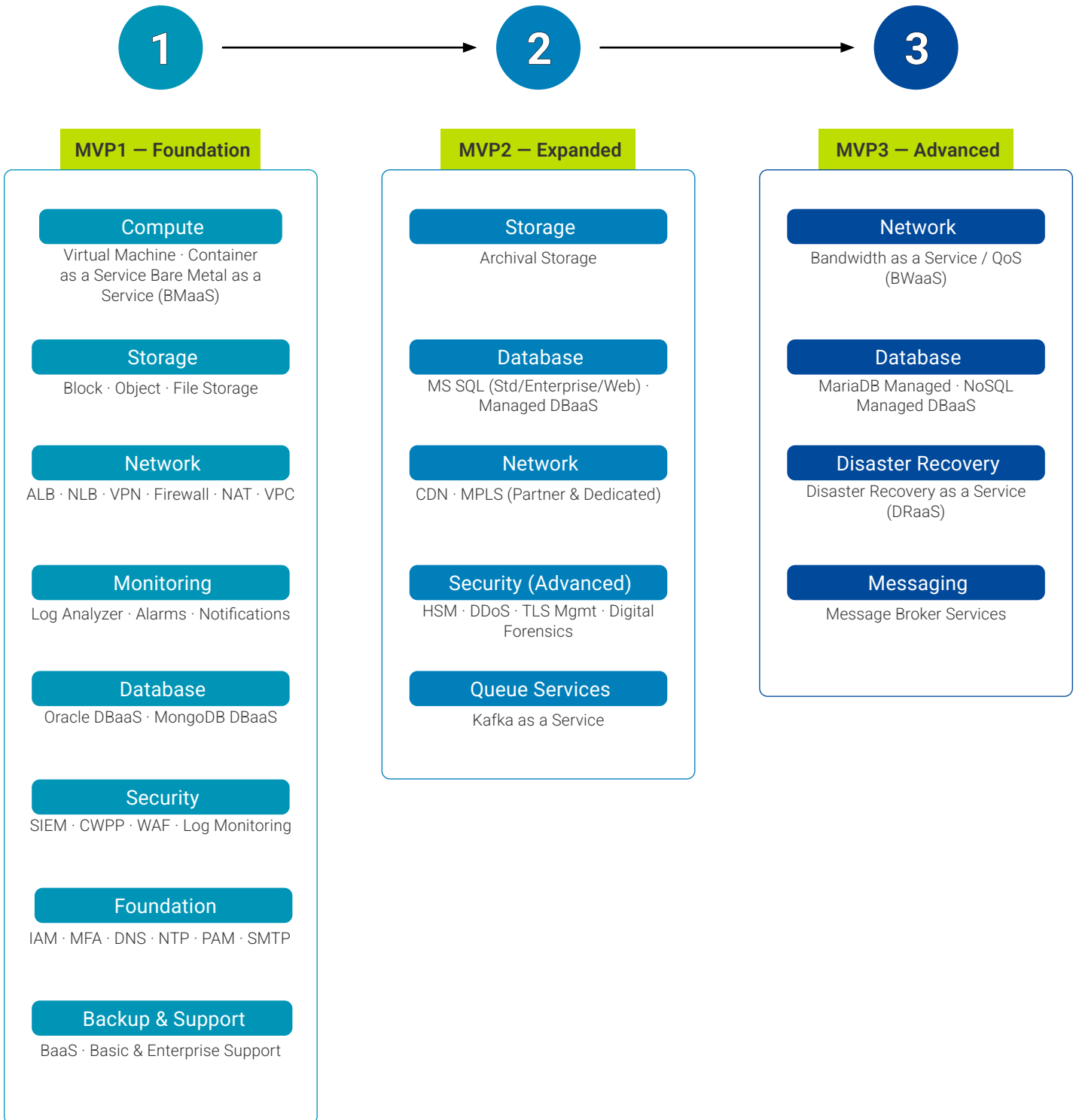


Figure 3: Phased Service Portfolio Delivery – MVP1, MVP2, MVP3

MVP1 — Foundation Services

The first milestone delivers the core infrastructure and platform services required for a fully operational sovereign cloud platform:

Category	Services
Compute	Virtual Machine · Container as a Service · Bare Metal as a Service (BMaaS)
Storage	Block Storage · Object Storage · File Storage
Network	Application Load Balancer · Network Load Balancer · VPN Gateway (Site-to-Site & Point-to-Site) · Firewall · Public IP · NAT Gateway · VPC
Monitoring	Log Analyzer · Operational Metric Collection · Alarm Service · Notification Service
Database	Managed Database as a Service (Oracle and MongoDB)
Security	Security Incident and Event Management · Log Monitoring · Cloud Workload Protection · Web Application Firewall
Foundation	IAM · SMTP · Identity Federation · MFA · DNS · NTP · Privileged Access Management · IP Address Management · Active Directory Services
Support	Basic Support Services · Enterprise Support Services
Managed Services	Managed Services
Backup	Backup as a Service

MVP2 — Expanded Services

The second milestone expands the portfolio with advanced storage, database, network, and security services:

Category	Services
Storage	Archival Storage
Database	Microsoft SQL-as-a-Service (Standard, Enterprise, Web Editions) · Managed Database as a Service · Database Licenses
Network	Content Delivery Network · MPLS Connectivity (Partner & Dedicated Interconnect)
Security	Cloud-Based Hardware Security Module · DDoS Protection · TLS/SSL Certificate Management · Encryption Services · Digital Forensics
Additional	Queue Services (Kafka as a Service)

MVP3 – Advanced Services

The third milestone delivers advanced network, database, disaster recovery, and messaging services:

Category	Services
Network	Bandwidth as a Service / QoS (BWaaS)
Database	Managed Database as a Service – MariaDB · Managed Database as a Service – NoSQL
Disaster Recovery	Disaster Recovery as a Service (DRaaS)
Additional	Message Broker Services

NOTE

The above list of services may change in accordance with the guidance provided by the Client Business team.

Architecture and High Availability

The CCP architecture is designed from the ground up for high availability, fault tolerance, and multi-region resilience. The platform operates across multiple regions and availability zones, with every layer of the stack engineered for continuous operation even in the event of infrastructure failures.

Multi-Region, Multi-AZ High Availability Architecture

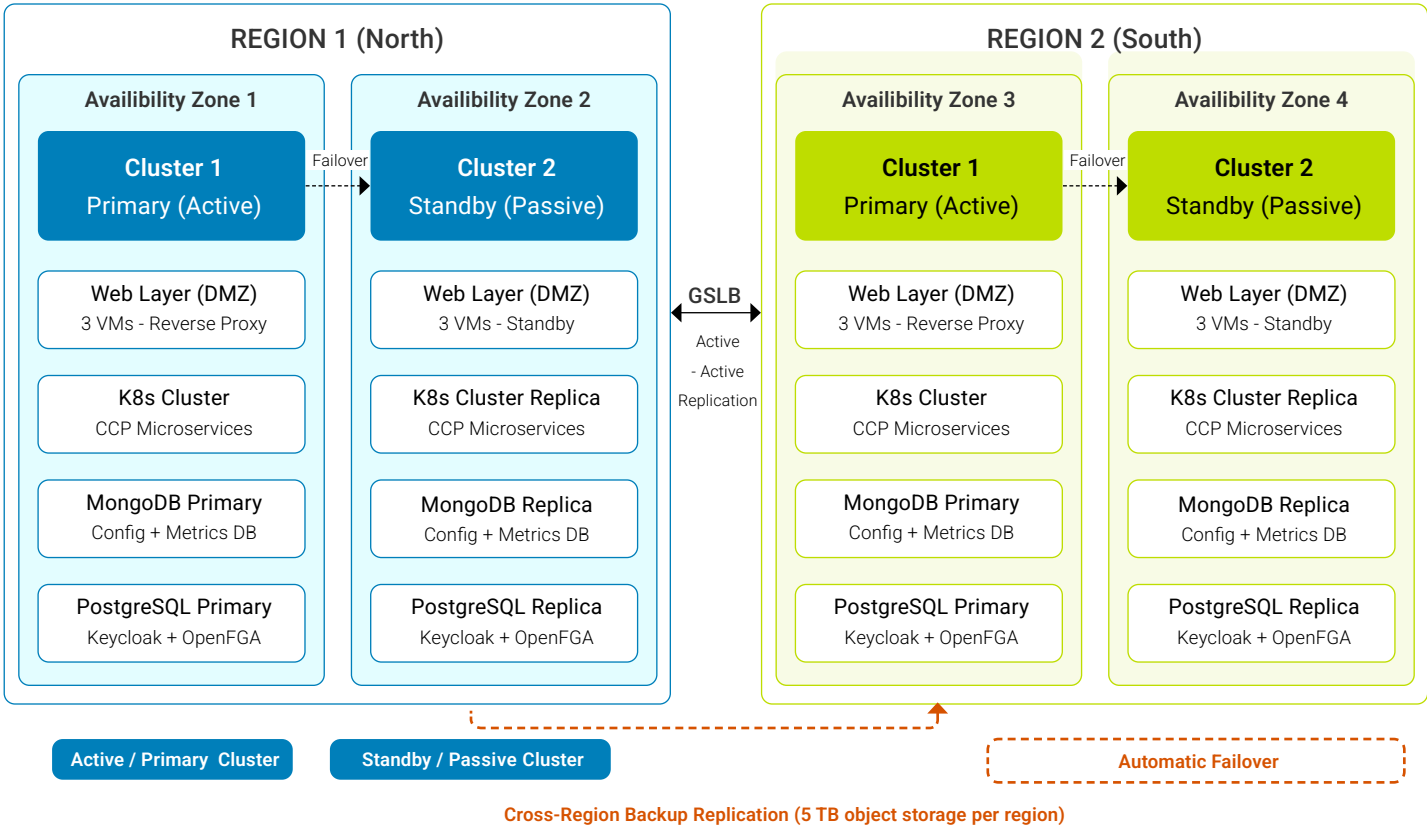


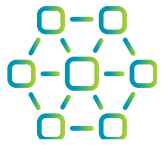
Figure 4: Multi-Region, Multi-AZ High Availability Architecture

Each region consists of multiple Availability Zones (AZs). Within each region, CCP runs independent components per AZ — all microservices managing infrastructure in that AZ operate autonomously. Two clusters run per region in an active-passive configuration:



Cluster 1 (Primary)

Hosts the main application services and primary MongoDB database in Availability Zone 1. This is an active cluster during normal operations. The web layer is deployed across 3 virtual machines in the DMZ, acting as a reverse proxy to the Kubernetes cluster in the production zone.



Cluster 2 (Standby)

Hosts replica application services and a replica MongoDB database in Availability Zone 2. This cluster remains in standby, ready to assume primary responsibilities immediately upon failure of Cluster 1.

Failover and Continuity

CCP provides automated failovers at multiple levels:



Regional Failover: Traffic is automatically routed to the passive cluster when the active cluster fails. MongoDB replica sets ensure data consistency during failover within a region.



Multi-AZ Failover: If an individual AZ fails, services fail over within the region without impacting overall platform operations. Load balancers and DNS routing ensure seamless traffic redirection to active services.



Global Services Continuity: Global services — responsible for organisation onboarding, metadata management, metering aggregation, and quota management — run in active-passive mode across regions. A GSLB probe detects endpoint availability and redirects traffic to the backup region when the active cluster goes down. Internal quorum is based on a $2n+1$ system for accurate active cluster identification.

Global Database Architecture

Global services store Tenant, Project, and User information in MongoDB, replicated using Active-Active replication with change-stream. OpenFGA (the AuthZ engine) and its PostgreSQL backend run in Active-Passive mode between regions, with writes going to the primary region by default since this is a read-heavy database. A 3+3 node setup

distributes database responsibility evenly across two availability zones. In the event of an AZ failure, the surviving AZ retains a full set of 3 nodes, enabling safe manual failover. Administrators can force reconfiguration — such as reinitiating leader elections in the surviving AZ when quorum is lost.

Backup and Data Protection

CCP implements a comprehensive, automated backup strategy:



Application Data Backup: Active CCP cluster data — including Keycloak PostgreSQL DB, Config MongoDB, Metrics MongoDB, and ETCD DB of the Kubernetes cluster — is continuously backed up to a geo-replicated object storage bucket. North region backups are stored in the south region and vice versa.



Database VM Backup: Database clusters hosted on virtual machines are backed up using the Veritas backup agent. Backup Schedule: Incremental backup every 30 minutes, full back up every 24 hours, with a 3-month retention period.



Storage Allocation: 5 TB object storage per region for configuration replication, 5 TB for log retention, and 10 TB cross-region object storage for backup data.

Identity, Access Management and Multi-Tenancy

CCP is built as a multi-tenant platform from the ground up. Each customer organisation operates within a completely isolated identity and resource boundary, governed by a structured hierarchy and role-based access control framework.

Platform Hierarchy

Resources in CCP are organized in a structured hierarchy that maps directly to the customer's BSS Portal construct:

Platform Resource Hierarchy – BSS Portal to CCP Mapping

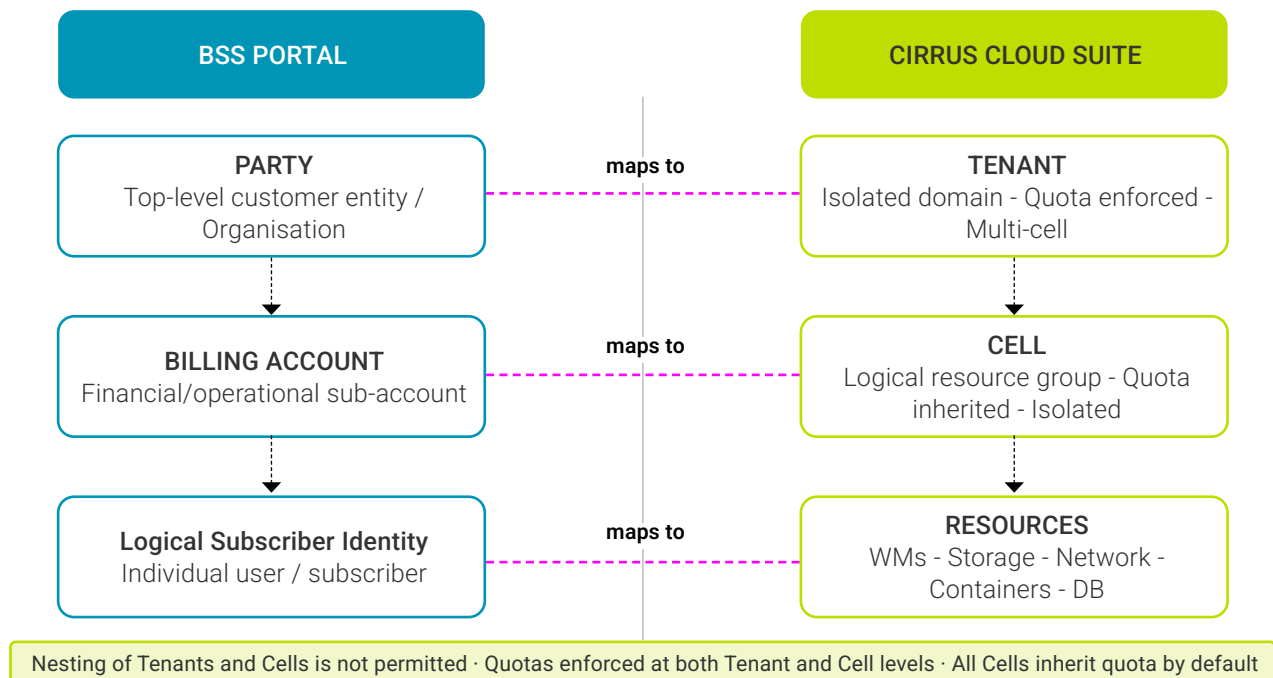
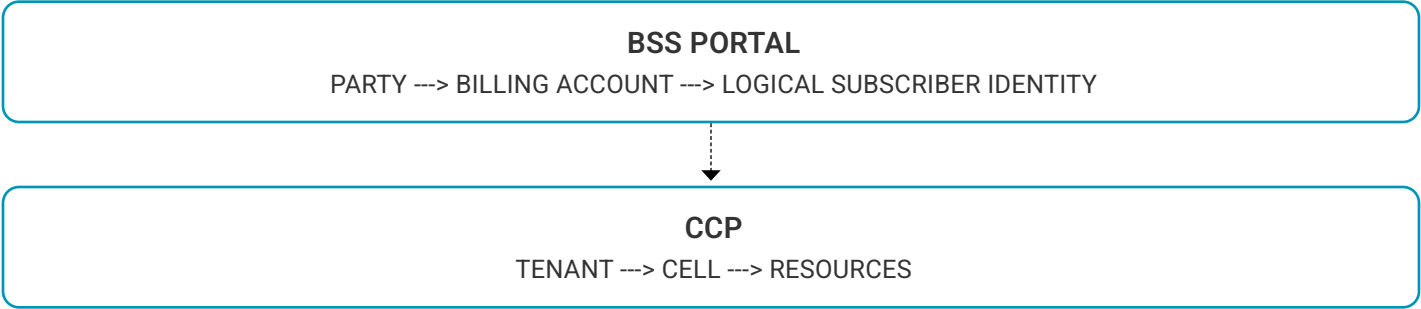


Figure 5: Platform Resource Hierarchy – BSS Portal to CCP Mapping



Each customer account maps a single Tenant in CCP. Multiple Cells can be created within a Tenant, providing logical isolation for different business units, projects, or

environments. Nesting of Tenants and Cells is not permitted. Quotas are enforced at both tenant and cell levels, with all cells inheriting quota by default.

Identity Federation

The BSS Portal serves as the primary user identity store. All customer user accounts are created, modified, and deleted exclusively through the BSS Portal. CCP IAM (Keycloak) federates with the BSS Portal as its Identity Provider,

supporting SAML 2.0 and ADFS federation. For each customer organisation, a unique Keycloak realm is created, ensuring complete identity isolation between tenants.

Role-Based Access Control

CCP provides a comprehensive, pre-defined role framework covering all platform functions. Seven organisation-level roles govern Tenant and Cell management:

Role	Access Level
Tenant Super Administrator	Root-level access; manages everything within a Tenant; can create other Tenant Super Administrators and Tenant Administrators
Tenant Administrator	Highest privileges per tenant; creates Cells and custom roles; manages quota and access requests
Tenant Viewer	Read-only access for auditing, compliance, and training purposes
Tenant Billing Admin	Access to quota usage, metering, and showback
Cell Administrator	Full access to all resources within a Cell; manages Cell quota
Cell Viewer	Read-only access to Cell resources for auditing and compliance
Cell User	Access to all services within a Cell; no quota management access

Eleven service-specific roles provide granular access control for individual cloud services, including VM Admin, VM Reader, Block/Object/File Storage Admin, Backup Admin, Network Admin, Container Admin, BareMetal Admin, Database Admin, and InfoSec Admin.

Scalability

CCP is engineered to scale with the demands of a sovereign cloud platform serving government and enterprise customers on national scale. The platform's Kubernetes-based deployment model enables horizontal scaling of all CCP microservices by adding worker nodes to the management cluster.

NOTE

CCP scales to support 50,000 virtual machines and 200,000 pods per deployment — with additional worker nodes added on demand to handle increased load.

The production infrastructure per availability zone is dimensioned to handle this scale:

Role	Count	vCPU/Node	Memory/Node
K8s Master Nodes	3	12 vCPU	24 GB
K8s Worker Nodes	5	48 vCPU	128 GB
CCP PostgreSQL DB	3	32 vCPU	64 GB
CCP MongoDB	3	24 vCPU	48 GB
OpenFGA PostgreSQL DB **	3	24 vCPU	48 GB
Global MongoDB **	3	12 vCPU	24 GB
Web Layer (DMZ)	3	12 vCPU	24 GB

** OpenFGA PostgreSQL DB and Global MongoDB VMs are stretched across 2 AZs in the region and routed accordingly.

NOTE

All cluster VMs must be deployed with anti-affinity enabled across different physical compute nodes to ensure cluster high availability.

Implementation Approach

CCP is deployed on a Kubernetes management cluster, separate from the workload infrastructure. The Cloud Management Platform solutions: Cirrus Cloud Platform, Cirrus Cloud Platform, and Cloud Orbiter — is deployed in the control plane of each availability zone and must not be deployed in the workload pod. This separation ensures that the management layer is not affected by workload activity.

Deployment Pre-Requisites

The following infrastructure and services must be in place before CCP deployment can commence:



Wildcard SSL certificates for CCP hosting and dynamic customer account URLs



Kubernetes-compliant storage with High IOPS performance



Load Balancer and VIPs for each CCP endpoint



Connectivity and credentials for SMTP server for email notifications



DNS Server with credentials to create dynamic domains based on customer accounts



NTP and DNS server connectivity



Accessible Container Registry to store container images



Connectivity and APIs to integrate with the BSS Portal

Phased Delivery

CCP services are delivered in three phases – MVP1, MVP2, and MVP3 – allowing the platform to be operational with foundational services quickly, while the full-service portfolio is built out progressively. This phased approach reduces

risk, allows for early customer onboarding, and provides the operations team time to build operational maturity with each new service category before the next phase begins.

What is Out of Scope

The following are explicitly out of scope for the CCP platform delivery:



Hardware procurement and its deployment



Performance Testing for components other than CCP



Software procurement and associated licensing (operating system, database, backup software, management software) other than CCP



Day 2 operations for underlying infrastructure (Compute, Storage, and Network)



Penetration Testing



Application or configuration changes within the BSS Portal

Conclusion

Sovereign cloud is no longer a future aspiration for governments and regulated enterprises — it is an immediate operational requirement. Building it successfully demands a Cloud Management Platform that combines the self-service convenience of public cloud with the control, governance, and security posture that sovereign deployments demand. Cirrus Cloud Platform delivers exactly this. With its unified management layer spanning virtual machines, containers, and bare metal; its multi-region, multi-AZ high availability architecture; its comprehensive role-based access control

and identity federation capabilities; and its phased, risk-managed service delivery model, CCP provides a complete, proven foundation for sovereign cloud platform operations. The platform's ability to scale to 50,000 virtual machines and 200,000 pods, combined with its enterprise-grade backup, security, and monitoring capabilities, positions it as a long-term platform for cloud services growth — from foundational IaaS in MVP1 through advanced database, security, and disaster recovery services in MVP2 and MVP3.

About CoreEdge

CoreEdge is the developer of Cirrus Cloud Platform, Cirrus Cloud Platform, and Cloud Orbiter — a complete, integrated Cloud Management and Orchestration platform designed for service providers, enterprises, and government organisations building and operating sovereign and private cloud infrastructure. Cirrus Cloud Platform provides hyper-

scaler grade Cloud Management Platform capabilities for OpenStack and Kubernetes-based deployments, enabling service providers to deliver IaaS, PaaS, and SaaS services through a unified, self-service portal with enterprise-grade governance, security, and operational tools.

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